ENEE3522 – Quiz #3 with Solutions

Problem #1

A single phase transmission line uses 4 ACSR "*Leghorn*" cables in a 6" square bundle in one conductor and 2 ACSR "*Ostrich*" cables with 9" spacing in the other conductor. The cable data is given below.

Calculate the **distributed series resistance** of the transmission line.

$$R = \frac{R_{sx}}{n_x} + \frac{R_{sx}}{n_x} = \frac{0.4319}{4} + \frac{0.3070}{2} = 0.2615\ \Omega / mile$$

Calculate the **geometric mean radius** of the conductor that uses the 4 ACSR "*Leghorn*" cables.

$$GMR_x = \sqrt[4]{\sqrt{2} \cdot \mathrm{d}^3 \cdot GMR_{sx}} = \sqrt[4]{\sqrt{2}\left(\tfrac{6}{12}\right)^3 (0.0172)} = 0.2348\ ft$$

Calculate the **modified geometric mean radius** of the conductor that uses the 2 ACSR "*Ostrich*" cables.

$$mGMR_y = \sqrt{\mathrm{d} \cdot r_{sy}} = \sqrt{\left(\tfrac{9}{12}\right)(0.0283)} = 0.1457\ ft$$

Problem #2

A three phase transmission line uses 3 ACSR "*Bluebird*" cables per phase in 10" equilateral bundles. The cable data is given below.

If the transmission line is 75 miles in length, calculate the **total per phase series resistance** of the transmission line **for a temperature of -5ºC**.

$$R_T = R\ell = \left(\frac{R_s}{n}\right)\ell = \left[\frac{(0.0476\ \Omega / mi) + (0.00019)(-5-20)}{3}\right](75\ mi) = 1.07125\ \Omega$$

Calculate the **geometric mean radius** of one of the phase conductors.

$$GMR = \sqrt[3]{\mathrm{d}^2 \cdot GMR_s} = \sqrt[3]{\left(\tfrac{10}{12}\right)^2 (0.0586)} = 0.34396\ ft$$

Problem #3

A single phase transmission line has a distributed series resistance of 0.054Ω/mile, a distributed series inductance of 2.86mH/mile, a distributed shunt conductance of 5μ℧/mile, and a distributed shunt capacitance of 10.33nF/mile. Calculate the **characteristic impedance** for this transmission line.

$$R = 0.054\ \Omega / mi, \qquad L = 2.86\ mH / mi, \qquad G = 5\ \mu℧ / mi, \qquad C = 10.33\ nF / mi$$

$$\bar{\mathfrak{z}} = R + j120\pi L = 1.07955\angle 87.133°\ \Omega / mi, \qquad \bar{y} = G + j\omega C = 6.33764\times 10^{-6} \angle 37.914°\ ℧ / mi$$

$$\bar{Z}_c = \sqrt{\bar{\mathfrak{z}} / \bar{y}} = 412.72113\angle 24.610°\ \Omega$$

Problem #4

A 75 mile three phase transmission line has a per phase distributed series resistance of 0.054Ω/mile, a per phase geometric mean radius of 0.117ft, a per phase modified geometric mean radius of 0.132ft, and a geometric mean distance between phase conductors of 15.255ft. Calculate the **pi model circuit parameters** for this transmission line.

$$R = 0.054\ \Omega / mi, \qquad GMR = 0.117\ ft, \qquad mGMR = 0.132\ ft, \qquad GMD_{eq} = 15.255\ ft$$

$$L = k_{L3} \cdot \ln\left(\frac{GMD_{eq}}{GMR}\right) = 1.5677\ mH / mi, \qquad C = k_{C3} \Big/ \ln\left(\frac{GMD_{eq}}{mGMR}\right) = 18.8494\ nF / mi$$

$$\bar{\mathfrak{z}} = R + j120\pi L = 0.593455\angle 84.779°\ \Omega / mi, \qquad \bar{y} = G + j\omega C = 7.1060\times 10^{-6} \angle 90°\ ℧ / mi$$

$$\bar{Z}_{TL} = \bar{\mathfrak{z}}\ell = 44.50914\angle 84.779°\ \Omega, \qquad \bar{Y}_{TL} = \tfrac{1}{2}\bar{y}\ell = 2.6648\times 10^{-4} \angle 90°\ ℧$$